

ActInf Textbook Group ~ Cohort 6 ~ Session 18

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okay we're in cohort sixo question mark And discussing some applying active inference questions and Jeff has turned us to the inspiration topic so where were we Jeff or continue yeah so um several of us Dean tickles and Anna magdalena herado and um um Hector Manriquez we've been going back and forth for for about two weeks on on what inspiration means uh through the lens of active inference and and for me inspiration and I think this is a really good um definition and I think Dean offered this up and it was um what happens when you have a new new model or a new artifact that has no through line no uh contiguity for how you receive how you arrived at that specific phenomena or or or experience or or bit of information right it is it is that that is seemingly without cause Okay lot of ways we can model it just are there any more General comments or or definitions or or thoughts and then otherwise we're g to kind of walk this pseudo chapter 6 path for H how we would move towards structuring it you're basically asking like how does like Insight arise into like the shape of some like object you're looking at or some insight you're looking at is that what that last question Zach yeah this uh this reminds me of really core active entrance right whenever you're talking about like if you just have a whole bunch of particles you can differentiate the states of the particles are in by the states that seem to have this kind of sub thing going on and the maintenance of the sub thing and this thing want by virtue of existing very core definition of active inference right or I'm sorry the uh I'm forgetting the word of it though it's a very important word the idea that because it exists we can talk about the Dynamics of it existing I'm sorry I can rephrase that if you I did not that very conditioning conditioning modeling of something on that thing being like existing or measured yes absolutely and so it's going to maintain its own state but the question I have is could you guys describe the no three line or continuity because by saying that it is a thing aren't we already talking about it has a thingness about it we are already talking about some states where it is instead of some states where it is not yeah like even a really in inspired new line of poetry it's like it it may have a chain of

causation so then if we were if we were um tasked to make an active inference model um and in that first question chapter six like what system are we modeling it's like what what question are we trying to explore are we trying to reify our definition or observation that inspiration is that that arrives at with without uh uh cause or are we we looking to quantify like some index of inspiration where on the lower end there's like complete continuity and then on the upper end there's less continuity here's one Boulder paper with the kind of more on the the Insight [Music] um okay so from chapter six we have those four stages of the question let's just answer the first question and then we'll look over to to a few other um kind of uh tools for modeling but what what system are we going to model like can we pick some specific inspiration setting just to make it clearer a song comes to an artist in a dream okay here I'll I'll I'm going to pop over to this other um active block fence Koda just because um it it complements the chapter six so in the textbook chapter six gives these four steps um that suffice to design active inference generative model so big picture the first part is like system of Interest identification systems Engineering Systems modeling then you get into the kind of chapter 4 chapter seven and eight questions like is it going to be a discrete time or continuous time what kinds of State spaces are the variables and then question three and four are about setting up basically the agents and the niche and it's this is dealt with at a pretty overview level in the textbook and um depending on how how complicated the setting is that this can be like very straightforward like thermostat and the room or this might be really uh sophisticated like having a bunch of Nest mates in in a colony but here here's just a few um compliments to those lists here's from the blockfront side um the cognitive modeling side first off the settings many discussions over the years and and sub discussions and things like that that that people have brought up but just focusing on the recipe um AI um this kind of just gives a little bit of a of a different flavor which is first take take a structural approach to the project like it asks a few more background questions about what what is being addressed and not then this was like some some of the earliest um memes with yakob which was like we're we map building with these cognitive models and then we're playing with this like der because you could derive something is kind of like to to squeeze out but then also it means to take the derivative and then there's different notations for the derivative like sometimes it's a DOT or it's a prime or it's like f fine of x so then if we derive a model but then there's different notations for the model and then the table that I'll copy back into our um textbook group this I'll just take it from the ant model okay here are active inference ontology terms I'll delete these columns and their um notation so we see a b CDE e pi a o s and then here we'll we'll do artist generative

model once we fill out these cells for like if we're going to use a discrete Time model um then we'll basically have all the variables we need but this is just a way to use the ontology to scaffold making a generative model because basically you just fill out what would these things be for that situation and then already just manually or with a llm there's enough information to get you to the PDP okay what should the observations be not the components of the song I suppose yeah this also this is like part of that beginning um going from the qualitative and starting to give some structure it's like is it going to be just a sound model and then H how many layers to go into detail but that's kind of that that journey is to take the the um pre structured situation like just situation as it is or or or or um uh not explicitly categorized situation and then be like what are we going to pull out as observations for whom so it's like like will there be a bodily awareness will there be a time awareness if we're going to ask a question about time then I mean we may need that to be explicitly included or but maybe it's not in the observations let's start with just music sound and we can figure out what types okay what hidden states are being inferred so what are what are um unobserved but hidden causal factors that actually exist or are proposed to exist if there's lyrics then you have to suppose the understanding of the language same with instruments right you have to have a model of of which sounds or at least how you can instrument the sounds that you yeah how you can instrument the sounds there's also an interesting piece there with this um if we're going to come back to the time perception because music has a Time rate but like a architectural Vision or something that might be static or so then but the song I mean does the artist hear the song at Onex speed or do they hear it like faster or slower and then and then what if it's um not possible to be played or do they get it partially and then fill the rest of it in when they wake up yeah true underlying song like I it like it sounded like they were underwater but I heard something okay let's just also just to be clear with the letters and I'll just bring it over okay so we have some observations coming in at the very least music possibly some time observations or some rhythm in the S we have two two kinds of s's States related to the music and then States related to the flow of time okay what does that give for the a matrix then the the simplest answer is a is mapping between s and O what whatever those are so that could be pure intra domain so that would be like sound o maps to sound s time o maps to Time s so in that situation there' be like two separate s um variables they could be different shapes and then each of them has an a leading to separate o variables or there can be more of a domain Fusion like um single um s with mapping to sound o and sound and time o me this is like why it's just like get it down and like kind of iterate over it it's like oh because like what is

what is the really the time observations like well is is that an internal observation is that a metronome is that a breathing so it's it's like we can kind of run back over and check for for a consistency but basically a is going to do whatever intermediation work is required to get from o to s but this is the sense making partially observable element that's going to go from whatever you're saying is data as measured to variables as proposed just thinking of like another ingredient like sometimes as a musician you can like feel the song in terms of like how you hear it more like through your spine or like energetically but then you you might also like that might get mapped to how it feels like to play the song you might have some kind of intuition of like the fretboard or whatever it is yeah that's good good idea like using um kind of like Intermodal her istics like um this cake is going to taste good because I just touched it and it's bouncy so then there could be in that situation that would be more like the domain Fusion for a because you'd have like the Latent state would be like quality of the cake and then it'd be like um here's how the color uh profile is associated with it and then here and then here's how the um touch here's how the springiness is associated with that latent variable of cake quality but then in the dark room you still could use the touch modality to assess the cake quality okay let's move Pi up so what actions are being modeled as being taken actions are going to intervene they're going to pick out a slice of B and then the chosen slice of B is going to be used to control the time evolution of s so again kind of an interesting case like what are the actions in this situation where um the inspiration is putting on a show for the artist what action selection are we looking at for them so I can get grounding we saying like the action is like basically like as the dreamer like as the person's like waking up from the dream and trying to like grasp onto the song and and solidify it or is this like a different kind of question we can make this really like non-physical right because it's coming from a dream therefore we can say well could we say that like one of the actions is changing our model of the world as in now I want to make this song before I did not want to but then I had the dream which is a perception and the action I take is the desire to make this thing in reality okay yeah let's let's get some of those so let's just say that one one kind of action would be if s equals motivation to make song would be um so so through time there's this motivation to make some song and then there's B that simple two two options for B either keep your motivation where it was or increase your motivation would be choosing to increase motivation obviously other ways to model that too here if s um Martin brought up this like waking up there's modeling of attention as action that's some of the work with um lar SED Smith where internal action policies covert action policies are modeled as attention so this is like s would be what am I paying attention to and then Pi is

a policy on what is being attended to but it's but but kind of using this skeleton like our um recipe outline it's like whatever whatever s is there's going to be some s's that don't have direct policy control those are called uncontrollable States like in pmdp um and then there are the s that are controllable which means that their bees their their their unfolding is controlled by bees that have some policy efficacy but then this this this dream um setting kind of um stretches the case because it's like is this a lucid dream where the person is seeking to um you know is it really them from behind the veil inspiring them but then obscuring that it was them but then this goes back to that first question of chapter six which is like which system is are we modeling and um inactive inference just like any anywhere else people have done systems modeling that's not necessarily trivial because once you start tracing the physical and the informational exchanges of a system you can you know you you tug on one part of the world and everything else comes along with it so that that does become the question like which modalities to restrict to on the other hand a great strength of active inference is that if the models are constructed appropriately at least they'll be interoperable like you can say we're making just the dream auditory track and then just the dream proprioception track and then you could test those two models separately and then you could more easily also fuse those models see if there was like um synergistic outcomes so there's still a lot of possible modeling but at least what you make maybe more interoperable let's see what other s could be get an idea of like the framing here maybe it's a question but um the pi is like some kind of like action that changes the state but does that also like create the flow of time in a sense that just how States change over time yeah great question just the the the narrower answer is pi is a probability distribution over policies policies are the affordances that can be taken in a given moment over the given time Horizon so if there's no planning then Pi are just the actions that can be taken in the next moment which are the affordances like up down left right if there is planning then Pi starts getting bigger like exponentially bigger with the time Horizon like two time steps um if there was four affordances there'd be 16 options in pi so Pi is a probability distribution it sums to one because some action is taken and then depending on which of those um indexed policies are selected that's which B transition operator is chosen and then that's what controls how s at T minus one goes to S of t Okay so but then when it comes to modeling it's like um and again time time perception is another another fun and interesting case but it's like whatever what are it doesn't reach in directly and change the temperature of the room it's like temperature of the room at the previous time and then you decide turn on the air conditioner or not and then if you choose turn on the air condition conditioner then you got the

prior to the temperature turn on the air conditioner and that results in the next temperature so it's not that that Pi directly like goes in and changes the temperature in the room but it acts through this Bayesian process yeah appreciate that where and there have been a few time perception active inference papers I'll try to find yeah we're definitely playing with the weird system because time perception and dream changes okay here's Bogata and Jaa I forget which number we can look but it was a live stream previously so here here they're using the haerian phenomenology ontology so retention is what is retained or sedimented from the past protention is what is anticipated into the future so this is kind of like the thickness of the presence where like what was what was uh anticipated moves through the the present into the um and then be parts of it becomes retained here okay this might help us see here okay font looks a little bit odd but this is this is State so this is basically this the the table don't know where the table were but this is basically what the table is and this like this is what applying active inference looks like in this setting where we're doing the ABCDE e o it's about defining what those variables look like for the given system or a question of Interest is it just me or does that look a lot like relativity except the gamma term has been replaced by a probability distribution which which part uh mostly that first graph that you showed where uh there was that kind of diagonal set of squares that's what really set it off for me uh and talking about uh the object time versus the model time uh using to and T that was another thing that made it really seem similar that one right there exactly so what what would this be like in um relativity that would be uh changing reference frames which would mean that like they are perceiving space and time differently than a different Observer would which is to say that like one person at point B would look at something and say oh uh maybe Point uh I'm sorry a top left d uh would look at something and be like oh I see two events in two different points in space they happen at time one and time two but at person's point AF of to the far left would look at those same two events and say no they didn't happen at time one and time two they happen at time three and time four they would happen at relatively different times because their experience of time would be slightly different oh yes I see this like or it could be like here's here's um message one two and three person a has gotten none of them person B has like yeah I got message one but but I'm just getting message two now but then for this person they were closest to this I don't know if it's exactly how it's laid out here but it's like already for them the messages are in the past but for someone else those messages haven't even happened yet yes exactly the time is relative okay we we'll see what we can bring back into the model but also looks kind of like a fretboard like a diagonal line on a fretboard for a base well it totally does

we we can make him um song compression okay that I mean was that just was that just an example of inspiration there it might have been I don't know it looks like it how much how and also this is kind of related to heck and Michael Walker's work on the side by side in the zone of of bounded surprisal which is like there's um the question of the the insight to One agent and then also there's the question of like whether it can be observed and whether that can be transmitted like the the just to make it more concrete there studying these flaked Rock artifacts that that um like sort of human ancestors made and then they're always like popping up but then they go away but like it will be it will have been made a certain way but then it's not transmitted how it was made so just because something can can can be seen by one doesn't make it seen by another let's just see if there's any other pieces to this paper then we'll return to where we were I mean the this is there's obviously a lot to it here and they're they're going into the they're kind of Bridging the H seran phenomenology with the future present and past with the active style but that's kind of the meta rest recipe which is combine the actim fonty the the letters and the functions and the methods that we know with some other domain time

Consciousness is kind of a a particularly there's it it has because the the um the measurements come through another modality there's not like approximate sense like you could have a a a vibrating clock or a metronome or breathing but then the time inference is something that's added on to some other modality whereas in sound it's like sound observation and then you're observing a sound latent state let's just pick one kind of policy to model if s were true song Tempo Pi would be speed up stay same let's just say that it's a elcid dreamer with with this limited kind of control over the music that they're hearing they can they can change the tempo so if if Pi has three elements for slow down stay the same or speed up up so this would just be like a simple Vector with three um cells if it were uh uniform it'd be like $\frac{1}{3}$ $\frac{1}{3}$ $\frac{1}{3}$ see these are preferences so does the person have a preference for the observations is that kind of like if the song seems Salient to them they'll try to like remember it or something like they just kind of reference of what the song feels like yeah I mean we it would have to be sketched out but what this this is going to go this is going to load on pragmatic value C is basically is is 100% where pragmatic value gets specified if C is flat then there's no pragmatic value in the generative model you only have epistemic value so it's kind of interesting like um is it is the song recognized as valuable because it's what they prefer or is it recognized as as insightful because it is recognized as novel something could be both novel and preferred however if if you recognized to the extent that you recognize it as part of your prior preference that would not be novel or if there's like notice that there's some kind of like Beauty to it and

the sense of like truth beauty and goodness like it's maybe some kind of meta structure that has some geometry to it yeah I mean this is why we're um why in the textbook it's like the teamas and the simpler cases because once we're dealing with a kind of um mental kernel that that has these um that for any input is going to start exploring like good true beautiful structured Etc then it's like um it kind of makes the sensory inputs just a small piece of the picture whereas in the textbook examples it's a very kind of closed form model where you can trace exactly how the observations relate to like changes in the model's Behavior so D this is the prior on hidden state so if we're going with s is the song Tempo let's just say prior on initial S so this is kind of like interesting um maybe even the song Tempo could relate to the time and the Rhythm so it's like if we're hearing sounds we hear one beat so but we haven't heard a second beat yet so we can't yet estimate how um how fast the song is so all you have is the prior on the song that's that would be all right when you're right here right yeah e this this is the Habit that habit this is prior on policy um it it's it's not it's it's right here e um so this is this could be uniform or not but basically just your your prior on policy and then a is is the action but we we it's the it's what policy selects okay then what about the generative process again this is a very interesting question where it can go like back and forth between sketching out a little bit about the agent then going back to the niche like so what is the niche of this artist that we've been sketching out what you mean by Niche like a musical Niche like the genres they like or something um I think it it could be um I'm just going to go to the the paper with the attention but like just just generically the observations that the artist is getting are coming from the niche and then the actions that the artist is emitting are being received by the niche so it's just that's when you plug in the agent model into the niche so that can be um like in the case of sensory data and action like bodily overt action that um is sometimes related to the physical Niche like it's getting thermometer observations from the room and then it's sending control signals to the air conditioner but then for example in this is the model where policy is is um here like here this let's look at this top level um attention let's look at okay just the two-Lev model OKAY the blue is the proximate sensory so this could be like visual data from the eye and then here is on the the orange levels nested model it has the same structure as as this as the PDP um except come in sa and then instead of Sao instead of like um what shape is or what color am I seeing and then what are the raw pixel data coming in that's the primary approximate visual layer but now here the SA but then the observation that this higher order agent basically this metacognitive agent is receiving is related to the Precision or attention on the lower level perceptual task so from the point of view of this orange agent its generative

process its Niche is the blue agent the blue agent Niche could be considered to be like the outside world but that's where you actually get to plugging in essentially systems across the interface everything about the generative model like defining the agents generative model is is defining your your um race car and then you have the track so you can generate a a really sophisticated model but essentially it's just going to stand freely until you close the loop with something even if it's trivial like the environment just um hands out random observations and it just soaks up and absorbs the actions the actions have no efficacy you could do that then then the agent's generative model policy selection doesn't matter right so the the context within a dream often just gets like rapidly moved around so the environment of this like internal representational world is kind of always changing but then if you're in the process of waking up some looks like it's growing you know back into Baseline reality you like slowly P open up different doors of perception right h I I I like the doors of perception uh connection and also makes me think like when things are quote normal seeming it's like me looking at my desk and and uh you know seeing a little flashlight it's like well that didn't take inspiration because the flashlights actually there and then it's like but then if it were something Fantastical then that but then it's then it's kind of like but if it was Fantastical and uninvited then that's a very fine line with like is that inspiration or is it just delusional if it's something that's that's not that's not that others don't see on the desk it's like okay inspiration but then is inspiration always associated with a positive sentiment like motivation or like with the veillance or with a feeling like some something has been achieved or could be achieved or you know is is in is the is our definition of inspiration going to include like a persistent um anxiety inducing delusion yeah it's a good point like a hallucination we're not like converging on reality I think is what you're saying you guys have done these before I'm assuming I'm not usually you've done this like thought experiment before this model but I'm assuming it's not usually in such a abstract dream space concept applying active inference models in different settings like yes yeah yes definitely all all paste in the chat like if you want to see here several dozen code examples um these these um are written by many different authors and then what the textbook since 2022 and what this other um collaborative work has done has like helped us set up more of that recipe and and the the procedures around it so that like within one to few conversations across these different settings for cognitive modeling like we can actually move forward with a domain specific application which is kind of the goal of the textbook it just the textbook doesn't take one all the way there but that's that's the kernel and that's helping familiarize with the structure of the model capacities some kind of features and uh aspects of the model and

then anytime you apply it into a domain this is pretty much what it comes to look like you just okay here's the general model schema be like if we were doing a linear aggression on Healthcare be like okay observations are the healthcare data hidden state is the risk odds for this or that let's do the regression this is like that except there's this downstairs sense making component and the upstairs action selection component but yes we did CH choose a very um with time and then with this like with all these unknown unknowns where where are What is the nich is is it an is the dreamer creating a a mental Niche for themself or so some some useful elements but with some things that are simpler in if this had been a um robot moving on a chessboard it would have been a simpler model appreciate there this context it's cool yeah it's fun it is fun and and here's here's uh all here's uh yans he did this for his internship in The Institute but he used those those figures from lars's paper so this was just about time proximate perception sensory perception precision and then attention on time yielding a metacognitive estimate of how much time has passed it's really cool yeah it is really cool and then that's kind of again where the interoperability of these models can be can be really powerful let's just say like this is a very strong time perception or you know let's just say we have this and other time perception kernels then somebody on the music and aesthetic side has made a music then there's at least a path towards fusing the time and the Aesthetics model without needing to perhaps go to square zero for both of them I'll copy that into the thank you for that it also with the uh L paper that you were talking about earlier kind of makes me it leads into a deeper question that I have because I just don't feel like I have a solid understanding of how time is put into these models and the relativity thing makes me question how space is put into these models it makes me feel like there's going to be some kind of curis some kind kind of motif that models are generally going to have in order to move through their environments that we relate to what we perceive to be time or what we perceive to be space and so I think that there's going to be some relationship of variables that goes to that but whenever it comes to the hard definition of what is time whenever we're talking about this and I might be overthinking it but I think that there's still an open question there yeah I think I think so like for for um agents that work in time but don't necessarily have time awareness like if there was going to be a 100 robots and they were just unfolding in time but they didn't necessarily have to like bootstrap their own time concept that would be a simpler setting but then when when we start to kind of take the Chris Fields physics as information processing seriously and if we want to get to the first principles then we're dealing with agents constructing their own spatial temporal reference frames and then as the modeler there's that question of like how how does

the modeler or the model sew that fabric together like if if if two agents have different um one one thinks it's 2 o'clock one thinks it's five o'clock but then like is the is the modeler GNA just say okay the earlier one is right or we flip a coin to decide who's right it's like will those will they even be able to talk to each other and then if the modeler says well yeah it's 3 o' they're both wrong then then then what system are you modeling if you're modeling how time delusion can can um can coexist then that's one versus I don't know it it really comes down to what you're trying to model and and then more of the the research and speculation side might take us to what structure could the models have and then more on like the RX and for RX environment side we'll Explore More empirically like what are the nuts and bolts of getting of um different time agent agent specific time systems to to interact but that's that here it's this is just sort of the skeleton well okay facilitators time question mark um in the coming weeks will go to chapter eight continuous time so that'll actually be fun and uh exit the discreet into continuous time setting um chapter nine putting empirical data in rather than just generating synthetic data fitting with empirical data I hope by that point whether with pmdp or another method let's let's like get an example for that 10 a wrapup or if you are going to be coming to the chapter 7 then we have chapter three coming up the high road chapter four generative models and then chapter five neurobiology so cool thank you all thank you so much cool see you next time bye